

Development of indoor environmental quality index using a low-cost monitoring platform

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ABSTRACT

The fundamental goal of modern high-performance buildings is to achieve low environmental carbon footprint while securing indoor conditions that promote human health and well-being. Current procedures for indoor environmental quality (IEQ) assessment are often sporadic, expensive, intrusive and often limited to few parameters and experts-use only. This research examines an overall IEQ assessment model in three field studies. The assessment of IEQ is achieved with the methodology for the integration of relevant IEQ parameters into an overall index through derivation of their individual weighting coefficients. A low-cost open-source platform "ENVIRA" was developed and deployed for continuous measurement of relevant IEQ parameters. A total of 125 occupants participated in the field studies – 69 in two open offices and 56 in the educational building. The results from occupant survey regression analysis for offices showed that indoor air quality is the most critical IEQ parameter with the weight of 0.35 followed by thermal comfort (0.285), acoustic comfort (0.195) and illumination (0.17). The occupant preference differed in the educational building where thermal comfort was the most dominant component (0.31) followed by indoor air quality and acoustics (0.25), and lighting (0.19). Results show that individual weighting coefficients are not equivalent and could differ across building types. Finally, the mean absolute error between the average survey results and the overall IEQ index for the three field studies was <3% suggesting that the platform can be used for generating large IEQ datasets for researchers and practitioners.

1. Introduction

Indoor environmental quality (IEQ) can be broadly categorized as a multitude of indoor environmental components: thermal comfort, indoor air quality (IAQ), acoustics, visual comfort, interior layout, biophilia, building's location, and connection to services, etc. (Al Horr et al., 2016). However, only the first four environmental components are variables that can be physically monitored and that seem to be the most relevant to our health and productivity (Mujan et al., 2019).

Traditionally, scientists and professionals have evaluated IEQ using sparse scientific-grade equipment and meticulous measurement procedures (Heinzerling et al., 2013). High investment costs, and lack of ubiquitousness, accessibility and interpretation of the information that supports occupant needs, have deterred occupants from individual monitoring of IEQ parameters. With the recent emergence of low-cost (consumer-grade) Internet of Things (IoT) environmental monitors on

the market, the scientific community has shown an increased interest in their assessment. The term low-cost is used as a reference for consumer grade devices ranging in price up to 500 EUR on the market (Tiele et al., 2018). Currently, the majority of low-cost consumer grade devices are in the 200 to 350 EUR range (Demanega et al., 2021). However, the majority of the field and controlled evaluations have focused primarily on IAQ, and mostly on sensor detection capabilities for particle number concentration (Sousan et al., 2017). Zamora et al. (2020) performed a one-year evaluation of three low-cost environmental monitors, but reported the results for the PM_{2.5} sensors only. Other studies broadened the research scope by including PM₁₀ sensors (Singer and Delp, 2018; Wang et al., 2020). One recent study performed a holistic assessment of low-cost monitors that include not only particles, but also carbon dioxide, total volatile organic compounds, dry-bulb air temperature, and relative humidity (Demanega et al., 2021).

Currently, no consumer grade IEQ monitor can be found on the market. To be able to characterize the mutual need and importance of

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Nomenclature

I_{IEQ}	Indoor environmental quality index [%]	PPD_{IL}	environment [%]
I_{TC}	Thermal comfort quality component index [%]	PPD_{AC}	Predicted percentage of dissatisfied with visual comfort [%]
I_{IAQ}	Indoor air quality component index [%]		
I_{AC}	Indoor acoustics comfort component index [%]		
I_{IL}	Indoor visual comfort component index [%]	c_{CO_2}	Carbon dioxide concentration [ppm]
I_{CO_2}	Carbon dioxide subcomponent index [%]	c_{TVOC}	Total volatile organic compound concentration [$\mu\text{g}/\text{m}^3$]
I_{TVOC}	Total volatile organic compound subcomponent index [%]	$c_{PM_{2.5}}$	Carbon dioxide concentration [$\mu\text{g}/\text{m}^3$]
$I_{PM_{2.5}}$	Particular matter ($\leq 2.5 \mu\text{m}$) subcomponent index [%]	$PM_{2.5}$	Particular matter less than $2.5 \mu\text{m}$ in size
W_{TC}	Thermal comfort weighting coefficient [-]	$TVOC$	Total volatile organic compounds
W_{IAQ}	Indoor air quality weighting coefficient [-]	LX	Intensity of light on a horizontal plane [lux]
W_{AC}	Acoustical comfort weighting coefficient [-]	$Actual_{SL}$	Actual sound level [dBA]
W_{IL}	Visual comfort weighting coefficient [-]	$Design_{SL}$	Designed room sound level [dBA]
PMV	Predicted mean vote [%]	MRT	Mean radiant temperature [$^{\circ}\text{C}$]
PPD	Predicted percentage of dissatisfied with thermal	RH	Relative humidity [%]

the IEQ components through the development of individual weighting coefficients (Wei et al., 2020), it is important to be able to measure IEQ in a holistic manner. Research papers present several low-cost arduino-based devices currently in a stage of development (Parkinson et al., 2019a). Also, the development of IEQ monitors is a proof-of-concept and their field deployment and overall performance is largely unknown. Ali et al. (2016) created a multitude of inexpensive arduino-based sensors for IEQ monitoring. Jin et al. (2018) presented a mobile autonomous IEQ sensing station. Karami et al. (2018) developed a sophisticated IEQ monitoring station with integrated analytical software and mesh network capabilities. However, this station cannot be considered low-cost nor easy to use by non-scientific users in its current state. As the application of low-cost monitors is continuously expanding, several studies demonstrated how they can be used for spatiotemporal mapping of indoor environmental parameters (Li et al., 2018; Scarpa et al., 2017). The most promising consumer grade platform SAMBA was presented by Parkinson et al. (2019a, 2019b) with a custom hardware design. It has state-of-the-art capabilities, compact and user-friendly design, but it is currently not available on the market and has no price tag.

The main objective of this research was to advance the holistic IEQ assessment and the development of the IEQ weightings components. The goal was also to create a practical low-cost IEQ monitoring platform (ENVIRA) as there are currently none available on the market. The ENVIRA offers an easy-to-interpret IEQ information that empowers the end-user to interact with the environment, and provides a rich dataset for further use by researchers and professionals. The novelty of this research lies in the conceptualization of a real-time holistic IEQ monitoring system accessible and approachable by occupants.

First, the IEQ categories and their components are discussed in relation to development of a unified IEQ index (Section 2). Second, the insight into the ENVIRA monitoring system, its components, and calibration procedures are given (Section 3). Finally, the monitoring results from the three case studies (two open offices and one educational building) are presented in which the ENVIRA platform is deployed. The environmental data is compared to the subjective occupant perception of IEQ established with the use of an occupant survey (section 4). Finally, the results are discussed with their implications on the holistic approach to a unified IEQ index.

2. Selection of IEQ parameters

Standard EN 16798-1 (2019) will be used as the starting point for the development of the IEQ index (I_{IEQ}). The Standard defines the main parameters to be used as input for building energy calculation and evaluation of the overall indoor environment. Five parameters of IEQ are recognized: thermal environment, IAQ, humidity, lighting, and

noise. These five categories are recognized for their impact on occupant comfort satisfaction and building energy consumption. The second category (IEQ_{II}) represents a normal level of expectation, while the first category (IEQ_I) is expected in indoor environments designed for people with special requirements (e.g. infants, patients, and elderly). Lower categories of comfort are expected to be associated with decreased occupant satisfaction. However, all of the parameters must be within levels known not to lead to health risks. Table 1 summarizes the four standard design categories of IEQ expected by the occupants.

The ALDREN project (Alliance for Deep Energy Renovation) represents the latest stride from the EU to implement rapid existing building stock renovations to improve energy efficiency (Wargocki et al., 2019; Wei et al., 2020). The project adopted the category scheme from the EN 16798-1 and converted it into an ALDREN-TAIL Index. The purpose of this index is to take into account IEQ during renovation of existing building stock. The metric incorporates four indexes (in total 12 physical parameters): thermal comfort index (T), acoustics comfort index (A), indoor air quality index (I), and lighting comfort index (L). Each of the components can achieve a four-color score from green to red (high to low quality). The overall IEQ is indicated by roman numerals from I to IV (high to low quality), with IV being equal to the lowest quality of the four components to incentivize IEQ improvement. Table 2 gives a visual scoring system of the ALDREN-TAIL Index.

Nowadays, an overall building IEQ rating cannot be achieved with the use of green building certification systems as they do not credit the overall IEQ directly. Rather, four main parameters of IEQ are assessed separately. Research articles are the only way to gain insight into the current understanding of the overall IEQ rating methodology.

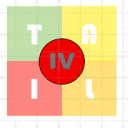
Chiang et al. (2001) first defined I_{IEQ} as a polynomial equation consisting of seven indoor environmental parameters with their respective weights. The study covered 28% of Taiwan's elderly care centers and used a consultative analytical hierarchy process to determine weightings. Wong et al. (2008) used a multivariate logistics regression model to find relative weightings from 293 surveys in office

Table 1

Categories of indoor environmental quality adopted from Standard EN 16798-1 and the proposed I_{IEQ} scoring system.

EN 16798-1		Proposed overall I_{IEQ} score (%)
Category	Level of expectation	
IEQ_I	High	$75 < I_{IEQ} \leq 100$
IEQ_{II}	Medium	$50 < I_{IEQ} \leq 75$
IEQ_{III}	Moderate	$25 < I_{IEQ} \leq 50$
IEQ_{IV}	Low	$0 \leq I_{IEQ} \leq 25$

Table 2
ALDREN TAIL index (Wargocki et al., 2019).

TAIL index	Component quality	Overall quality	Parameters	
	Green – high quality	I – high quality	Temperature	Formaldehyde
	Yellow – medium quality	II – medium quality	Sound level	Radon
	Orange – moderate quality	III – moderate quality	Vent. rate	Relative humidity
	Red – low quality	IV – low quality	CO ₂	Visible mold
			PM _{2.5}	Illuminance
			Benzene	Daylighting

buildings. Ncube and Riffat (2012) determined weighting factors based on two office buildings in the UK, and found similar results to those reported by Chiang et al. (2001). Piasecki et al. (2017) proposed an equal weighting coefficient system as each IEQ component is calculated by an equation determined from the results of the occupants' subjective perception of the environment. Recently, Buratti et al. (2018) investigated seven university classrooms and 928 survey participants to determine I_{IEQ} , however, their study omitted IAQ. Table 3 presents studies that reported weighting coefficients for IEQ parameters.

Existing studies typically adopt I_{IEQ} as an index having a linear relationship with its weighted components shown in Equation (1). The four environmental components are calculated from physical measurements of indoor environmental parameters. In practice, weighting coefficients W_i are most commonly derived from statistical regression results. The input data for the regression analysis are obtained by post occupancy evaluation questionnaire results. The occupants are surveyed on their overall IEQ satisfaction and the satisfaction with each IEQ parameter, thus giving weight to each IEQ parameter in their subjective perception of overall IEQ. In our study, the I_{IEQ} ranged from 0% to 100% of occupant satisfaction and was separated into categories as shown in Table 1.

$$I_{IEQ} = W_{TC} \cdot I_{TC} + W_{IAQ} \cdot I_{IAQ} + W_{AC} \cdot I_{AC} + W_{IL} \cdot I_{IL} \quad (1)$$

Table 3
IEQ weighting coefficients published in the research literature and derived from the requirements of several green building certification schemes.

Author	Number of surveyed occupants	Weighting coefficients			
		W _{TC}	W _{IAQ}	W _{AC}	W _{IL}
Chiang et al. (2001)	12 care centers for elderly in Taiwan	0.24	0.34	0.19	0.23
Wong et al. (2008)	293	0.31	0.25	0.24	0.19
Cao et al. (2012)	500	0.316	0.118	0.224	0.171
Ncube and Riffat (2012)	70	0.30	0.36	0.18	0.16
ASHRAE (2012)	52,980	0.12	0.2	0.39	0.29
Kim and Haberl (2012)	323	0.12	0.2	0.39	0.25
Mihai and Iordache (2016)	115	0.251	0.241	0.263	0.245
Fassio et al. (2014)	17	0.31	0.25	0.24	0.19
Piasecki et al. (2017)	No survey - author's proposal	0.25	0.25	0.25	0.25
Buratti et al. (2018)	928	0.35	–	0.35	0.30
Green building certification system		Relative IEQ categories allocation			
BREEAM (2017)		0.17	0.28	0.28	0.33
The U.S. Green Building Council (2019)		0.06	0.47	0.12	0.35
Deutsche Gesellschaft für Nachhaltiges Bauen (2007)		0.25	0.25	0.25	0.25

2.1. IEQ component models - I_{TC} , I_{IAQ} , I_{IL} , I_{AC}

The thermal comfort component is based on the ASHRAE 55 and ISO 7730 Standards, which predict the mean vote of occupant's satisfaction with the thermal environment in buildings with heating and cooling systems (ASHRAE, 2017; ISO, 2005). Standards adopt this model from Povl Ole Fanger, who researched the impact of the thermal environmental parameters on occupant satisfaction (Fanger, 1970). The predicted mean vote (PMV) model considers four environmental parameters (air temperature, air velocity, relative humidity, mean radiant temperature) and two user-related parameters (clothing insulation, metabolic rate). The calculation was adopted from ASHRAE 55 methodology for computer calculation, which corrected the ISO 7730 erroneous procedure.

PMV behaves similarly to an inverted Gaussian distribution when correlated to predict the percentage of dissatisfied (PPD) occupants. The calculation for PPD is found in Equation (2). After, the I_{TC} is calculated as a direct correlation to PPD. PPD is subtracted from perfect acceptance of thermal environment (100%), and a score is shown in Equation (3).

$$PPD = 100 - 95 \cdot e^{-(0.0353 \cdot PMV^4 + 0.2179 \cdot PMV^2)} \quad (2)$$

$$I_{TC} = 100 - PPD \quad (3)$$

Indoor pollutant sources, indoor activities, ventilation rates, as well as the outdoor environment can affect indoor air quality (WHO, 2010). The occupants' satisfaction can be directly correlated to physical, chemical, or biological characteristics of indoor air pollution. To derive the index value for IAQ, the main focus was on three commonly studied air pollutants: carbon dioxide, total volatile organic compounds (TVOCs), and particulate matter smaller than 2.5 μm (PM_{2.5}). These pollutants are among the most favored IAQ parameters by the majority of green building schemes and some standards as shown in Table 4. The lowest level of satisfaction with one of these parameters represents the I_{IAQ} , as shown in Equation (4).

$$I_{IAQ} = \min(I_{CO_2}, I_{TVOC}, I_{PM_{2.5}}) \quad (4)$$

Indexes for gaseous pollutants, namely I_{CO_2} , total volatile organic compounds (TVOC) I_{TVOC} and particulate matter $I_{PM_{2.5}}$ are derived from a concept of measuring sound in decibel, which is not an absolute unit (Jokl, 2000). Instead, it is a logarithmical change from measurement and reference.

The limits are set between the outside CO₂ concentrations of 415 ppm for unadapted occupants to the toxic levels of 15,000 ppm shown in Equation (5) (Hansen, 2017). Therefore, an indoor space with outdoor CO₂ concentrations would have a score of 100, while the toxic threshold would be scored 0. I_{CO_2} is directly correlated to CO₂ concentration in ppm (c_{CO_2}) shown in Equation (5).

Table 4
Selected IAQ parameters and their adoption by Standards and Green Building schemes.

Standard or green building scheme reference	IAQ Parameter		
	CO ₂	TVOC	PM _{2.5}
BES	x	x	
BREEAM		x	
DGNB		x	
EN 16798	x		x
HQE		x	x
KLIMA	x	x	
LEED	x	x	x
Level (s)	x		x
NABERS	x	x	
OsmoZ	x	x	x
WELL		x	x

$$I_{CO_2} = 100 - 70 \cdot \log \frac{c_{CO_2}}{415} \quad (5)$$

Similarly, TVOC are transferred to a logarithmic scale where the lower limit of $30 \mu\text{g}/\text{m}^3$ is used for perfect conditions (10% of the limit), while the upper limit is adopted from the [German federal environment agency \(2018\)](#) of $300 \mu\text{g}/\text{m}^3$. It represents a long-term average that should not be exceeded. Direct calculation of the TVOC component can be observed in Equation (6) where the c_{TVOC} represents the TVOC concentration in $\mu\text{g}/\text{m}^3$.

$$I_{TVOC} = 100 - 100 \cdot \log \frac{c_{TVOC}}{30} \quad (6)$$

Among particulate air pollutants, particulate matter smaller than $2.5 \mu\text{m}$ ($\text{PM}_{2.5}$) was chosen as a representative. As there are no existing guidelines for the indoor particulate matter exposures and there are no studies directly linking exposure to occupant satisfaction, the [EPA \(2012\)](#) AQI breakpoint scale was used to derive $I_{\text{PM}_{2.5}}$. The EPA scale ranges from the concentration of $0 \mu\text{g}/\text{m}^3$ to $500 \mu\text{g}/\text{m}^3$ as the upper limit for hazardous environments (24-h average). However, this index's upper limit was set at $150.4 \mu\text{g}/\text{m}^3$ as this represents the break-off point for the very unhealthy conditions to the general public. The $\text{PM}_{2.5}$ level of $10 \mu\text{g}/\text{m}^3$ was used as an upper criterium for good air quality ([WHO, 2010](#)). Component $I_{\text{PM}_{2.5}}$ was derived similarly as previous IAQ factors where $c_{\text{PM}_{2.5}}$ represents $\text{PM}_{2.5}$ concentrations in $\mu\text{g}/\text{m}^3$, according to Equation (7):

$$I_{\text{PM}_{2.5}} = 100 - 85 \cdot \log \frac{c_{\text{PM}_{2.5}}}{10} \quad (7)$$

The indoor lighting quality consists of many aspects like illuminance level, distribution of light, glare, daylight, luminaire characteristics, the directionality of light, etc. ([Kruisselbrink et al., 2018](#)). Certain tasks demand specific intensity of light on a horizontal plane, which standards acknowledge ([European committee for standardization, 2011](#); [Illuminating Engineering Society, 2013](#)). Several studies developed a regression model for occupant dissatisfaction (PPD_{IL}) based on the field studies in Hong Kong office buildings ([Mui and Wong, 2006](#); [Wong et al., 2008](#)). Equation (8) is adopted from those studies and is applicable for the light intensity in the range from 200 to 1600 lux.

$$\text{PPD}_{\text{IL}} = \frac{100}{1 + e^{(-1.017 + 0.00558 \cdot LX)}} \quad (8)$$

I_{IL} is estimated as:

$$I_{\text{IL}} = 100 - \text{PPD}_{\text{IL}} \quad (9)$$

As a growing concern, noise has been associated with harmful health effects and impaired productivity ([WHO, 2018](#)). Standard AS/NZS 2107 ([Architectural Joint Technical Committee AV-004 Acoustical, 2016](#)) was used as a reference to determine the dissatisfaction PPD_{AC} from background noise increase (A-weighted), which was further adapted by [Ncube and Riffat \(2012\)](#) in Equation (10). Both actual sound level ($\text{Actual}_{\text{SL}}$) and designed room sound level ($\text{Design}_{\text{SL}}$) are measured in decibel.

$$\text{PPD}_{\text{AC}} = 2(\text{Actual}_{\text{SL}} - \text{Design}_{\text{SL}}) \quad (10)$$

Component I_{AC} is then calculated in Equation (11).

$$I_{\text{AC}} = 100 - \text{PPD}_{\text{AC}} \quad (11)$$

3. Materials and methods

3.1. ENVIRA - A new IEQ sensing platform

For field measurements, a low-cost IEQ device was prototyped, developed, and named the "ENVIRA" platform. The ENVIRA prototype ([Fig. 1](#)) is compact, with dimensions $20 \times 25 \times 10 \text{ cm}$ (LxHxW) designed to be placed on the working plane. The body of the device is made from PLA plastic and was 3D printed as the original design. The design utilizes

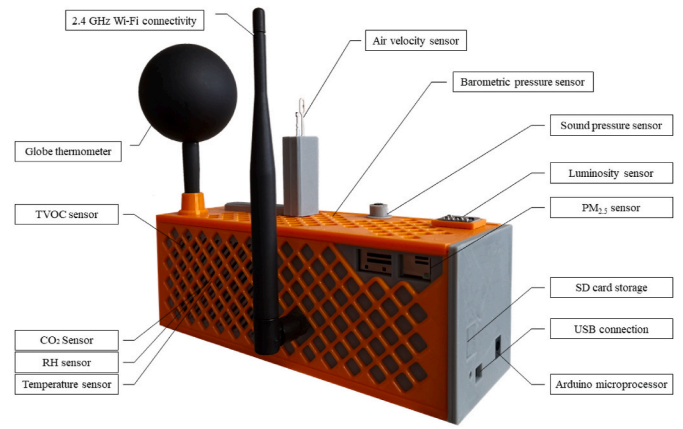


Fig. 1. Continuous IEQ monitoring device "ENVIRA".

as much natural ventilation as possible and minimizes self-heating while not allowing direct sunlight. The device is connected to a wall plug and powered with a minimum 12 V DC. In the future, autonomous operation will be enabled with the use of a battery pack. The retail price of all components is roughly 130 €, while the labor costs for design, calibration and manufacture are not estimated.

The ENVIRA platform is based on Arduino Mega 2560 Rev3 as it is currently the only board in Arduino series able to meet the multitude of requirements needed to operate the sensors. The variant of the board has integrated low-cost Wi-Fi microchip ESP8266. The ESP8266 enables the device to connect to any 2.4 GHz network. The microcontroller is programmed through an integrated development environment in C or C++. The board accepts different DC voltage inputs (7–12 V recommended) and has 5.5 and 3 V DC output, while one pin can be used directly without conversion. The total of 70 input pins (54 digital and 16 analog) provide sufficient sensor integration potential with I2C and UART serial communication protocols. More information about the Arduino platform is available online (www.arduino.cc).

The ENVIRA platform incorporates all sensors necessary to determine the I_{IEQ} , and records environmental parameters needed to calculate the individual index components I_{TC} , I_{IAQ} , I_{IL} , and I_{AC} . For thermal comfort, measurements for dry bulb temperature, black globe temperature, relative humidity, and horizontal air velocity are recorded. The Sensirion SCD30 sensor measures the dry bulb temperature and relative humidity. For horizontal air velocity, currently, the only viable option on the market is Wind Sensor Rev. P, as other suitable options would void the low-cost requirement. The body of the black globe thermometer is 10 cm in diameter and made from plastic. Inside, a 10 k NTC thermistor is used to measure the black globe temperature. Using thermocouples with an Arduino board is not viable as the sensor voltage outputs are too low to be read by the board, and an additional amplifier would be necessary. Mean radiant temperature (MRT) is calculated from the globe temperature. For the purpose of determining PMV and PPD values, metabolic rate and clothing insulation were estimated based on the field observations and literature ([ASHRAE, 2017](#)).

Among IAQ parameters, the ENVIRA is able to measure real-time concentrations of CO_2 , TVOC, and $\text{PM}_{2.5}$. In addition to air temperature and relative humidity, sensirion SCD30 utilizes NDIR technology to measure CO_2 levels. To make the measurements from SCD30 more accurate, a sensor for barometric pressure Bosch BME280 is incorporated into the platform. Notably, the Automatic Baseline Correction was turned off as it could cause performance degradation of similar sensors utilized indoors, which do not reach outside CO_2 concentration ([Barcelo-Ordinas et al., 2019](#)). Additionally, sensirion SGP30 is a hot-plate Metal Oxide (MO_x) sensor used to measure TVOC levels. Sensirion SPS30 is utilized for $\text{PM}_{2.5}$ measurements and uses the laser scattering measurement principle. The sensor was selected for its working

longevity (2 years with self-cleaning protocols) and compact size. The sensor can detect PM₁, PM_{2.5}, PM₄, and PM₁₀. During the testing and calibration of this device and others in the low-price range, it was determined that the technology is not sufficiently reliable for PM₁₀ measurements (Demanega et al., 2021). Thus, only PM_{2.5} (particles with diameter range 0.3–2.5 µm) was used as a relevant parameter.

Analog sensor DFROBOT SKU: SEN0232 records the A-weighted sound pressure level of the background noise to determine I_{AC} . The illumination on a working planar surface is measured by a light-to-digital photodiode converter Adafruit TSL2561.

Prior to ENVIRA's deployment, rigorous testing and calibration of each sensor against the high-grade scientific instruments was performed (Demanega et al., 2021). The overview of all sensors incorporated into the ENVIRA platform, including technical specifications, calibration values and reference equipment is summarized in Table 5. The measurements were set at 15s intervals for which the device was calibrated. In future deployment, interval (≥ 3 s) can be adjusted to suit the users' needs, however recalibration needs to be considered mainly due to CO₂ sensor self-heating (at 15s interval the mean error is +2.83 °C). All measurement records are stored locally on a microSD card as a CSV file and can be viewed live through the Arduino IDE monitor via USB connection. Measurements were averaged on a 1-min interval and sent to an open-source IoT platform Blynk (www.blynk.io), which reports the values in a real-time via a user-friendly smartphone application. Fig. 2 shows the custom Blynk smartphone user interface. On the main screen, component indexes and the IEQ index are displayed visually using gauges. The interface also contains detailed information on real-time numerical values of IEQ parameters (CO₂, sound level, illuminance, air velocity and barometric pressure) and their graphical representation (upper graph - dry bulb temperature, relative humidity and MRT; lower graph - PM_{2.5} and TVOC). Depending on the parameter values, Blynk allows messages to be sent to users including a daily measurement report via email.



Fig. 2. Blynk user interface.

printing house with the expectation of elevated concentration of TVOCs. The university amphitheater was selected as a comparison point to test whether interior space type plays a role in occupant acceptance of the four components of IEQ, and whether a uniform weighting scheme can be applied across various building types. All three field study buildings are visually presented in Fig. 3.

The occupant subjective perception of IEQ was evaluated by the use of a paper-based survey. The survey was handed out when occupants were acclimatized and at the same time when physical IEQ measurements with ENVIRA were performed. After brief instructions that explained the concept of the survey the participants independently filled in the questions. The general questions included information about occupants' sex, age group, and chronic conditions. Since the participants did not complete the survey simultaneously, they were asked to write the exact time upon completion. Consequently, a comparison between the real-time calculated IEQ index and overall occupant satisfaction could be made. Additional questions regarding the perception of IEQ components and IEQ as a whole were asked. A scale from 0 to 100 was used to determine the satisfaction level, with 0 being total dissatisfaction and 100 being total satisfaction. Thermal comfort was the only exception where the 7-point Likert scale was used ranging from -3 to +3, with zero denoting neutral state. The specific questions were:

1. In your opinion, what is your satisfaction with the current indoor environment quality?
2. What is your current thermal sensation?
3. How do you perceive the current quality of the air in the room?
4. How do you perceive current background noise?
5. How do you perceive current illuminance levels at your working desk?

The measurement of the first field study (FS1) took place on August 6, 2020. In the FS1, excellent IEQ levels were expected as the central

3.2. Field study buildings and occupant survey

In order to calculate the I_{IEQ} index (Equation (1)), a combination of field measurements and occupant surveys was deployed. Three case study buildings were chosen: two with open space offices and one amphitheater in a university building. Office buildings were selected on the basis of their mechanical ventilation system and their location - Novi Sad, Serbia. Also, the first site was chosen to determine whether occupants give more importance to acoustic components as the noise levels were reduced prior to tests with duct sound attenuators from 78 to 53 dB. The second case study was carried out in the offices connected to the

Table 5

ENVIRA sensor overview, measurement range, calibration accuracy and calibration instrumentation.

IEQ parameter	Company	Model	Range	Calibrated ^a accuracy	R ²	Reference instrument
Globe temperature	–	NTC Thermistor	–40 to 100 °C	±0.3 °C (10–35 °C)	0.99	Testo 480, K type thermocouple
Dry Bulb temperature	Sensirion	SCD30	–40 to 70 °C	±0.3 °C (10–35 °C)	0.99	Testo 435, Hot-Wire Anemometer
Relative humidity	Sensirion	SCD30	0–100%	±2%	0.98	LI-COR LI-850
Air velocity	Modern device	Rev. P	0–10 m/s	±5%	0.97	HOBO logger MX1102
Carbon dioxide	Sensirion	SCD30	400 to 10,000 ppm	±30 ppm ±3%	0.98	Testo 435, Hot-Wire Anemometer
TVOC	Sensirion	SGP30	0 to 60,000 ppb	–	^b	LI-COR LI-850
PM _{2.5}	Sensirion	SPS30	0 to 1000 µg/m ³	–	^b	GrayWolf AdvancedSense Pro - IQ-610
Sound pressure	DFROBOT	SKU:SEN0232	30 to 130 dBA	±1.5 dB	0.99	GRIMM - Model 1371 (miniWRAS)
Illuminance	Adafruit	TSL2561	0.1 to 40,000 lux	±3 lux	0.99	UNI-T UT352
Barometric pressure	Bosch	BME280	300 to 1100 hPa	±3 hPa	0.99	Testo Lux probe
						Testo 511

^a Accuracy evaluated after the initial 30-min warm-up of the unit sensors.

^b No calibration accuracy is given as it can be significantly influenced by an air pollution source (Demanega et al., 2021).

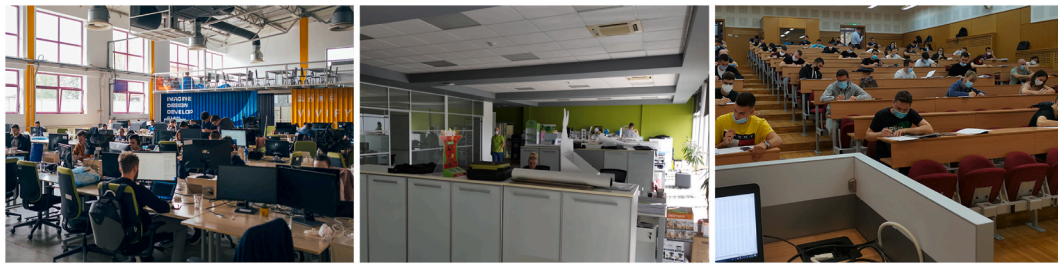


Fig. 3. Field study 1 (FS1) – Office building 1, Novi Sad Serbia (left); Field study 2 (FS2) – Office building 2, Novi Sad, Serbia (middle); Field study 3 (FS3) – Faculty of Technical Sciences, Novi Sad, Serbia (right).

HVAC unit was recently adapted, and the duct system underwent a partial reconstruction. The ENVIRA was placed in the center of the space at the working desk height (75 cm) and underneath the 600 mm air diffuser (5.4 m height) to record potential air movement. The measurements lasted continuously for 3 h between 11:00–14:00 h, during which the occupants filled out the survey about their environmental acceptance. The 34 employees were present at the time of the tests (22 male and 12 female) and all agreed to participate in the survey. 7 employees were between the age of 18 and 25, while the rest were not older than 35. All of the present employees were in good health, and none had chronic diseases. The metabolic rate was determined from the intensity of office work at 64 W/m^2 (1.1 Met) and clothing insulation from visual evaluation at 0.55 Clo.

The second field study (FS2) took place on August 26, 2020, with continuous IEQ measurements from 10:30–15:00 h. The ENVIRA was placed at an unoccupied working desk, with the same orientation as other desks (2 desks, 20 rows). 36 employees completed the survey (16 male and 20 female). One survey had to be omitted as the employee had chronic asthma. The workers were primarily in the 36–45 age group (97%). Clothing insulation was determined as 0.55 Clo, while the office work intensity was 70 W/m^2 (1.2 Met).

For both office buildings, the design noise level was set to 35 dB(A) as an A-weighted continuous equivalent sound pressure level (L_{eqA}) in accordance with [CEN \(2019\)](#). Some guidelines set higher design noise levels for open offices at 45 dB(A) ([ASHRAE, 2019](#)), while the former value was used as a stricter requirement.

The third measurement (FS3) took place in the amphitheater of the Faculty of Technical Sciences, Novi Sad, Serbia, on September 9, 2020, between 17:00–20:00 h. The ENVIRA was positioned in the center of the amphitheater, as well as in one corner of the space for the purpose of comparison. The amphitheater had no windows and relied on mechanical ventilation and artificial lighting. The 56 students and faculty staff filled in the survey while the continuous measurements were carried out. The leading age group was between 18 and 25 years of age (96%) and predominantly male (80%). All of the surveys were considered valid as none reported chronic diseases. 64 W/m^2 (1.1 Met) was assumed for writing activity and the average clothing insulation was estimated at 0.5 Clo. The WHO Guidelines for community noise provide 35 dBA as the acceptable noise level in school classrooms ([Berglund et al., 2000](#)). This value was used as a base of 100% satisfaction to calculate the acoustics index in the FS3.

3.3. Data analysis

SPSS Statistic software was used to determine weighting coefficients for components of the I_{IEQ} based on the questionnaire filled by the employees at the two offices and the students of one education building. Multiple linear regression was used as there is no multicollinearity between IEQ components, with the interception set at 0. This is logical as one would expect that if all of the contributing components I_{TC} , I_{IAQ} , I_{LL} , and I_{AC} are equal to zero, I_{IEQ} should also be equal to zero. If this is not implemented, the models for all three field studies show low probability

(p-value > 0.35) and an $R^2 < 0.45$. With the interception set at 0, adjusted $R^2 > 92.45\%$ and significance of $F < 0.0001$, which implies there is little to no possibility that all of the weighting coefficients could be zero and the null hypothesis can be rejected.

4. Results and discussion

4.1. Summary of descriptive IEQ data

The continuous 5-min averaged thermal comfort parameters for all three case study buildings are shown in [Fig. 4](#). While in the air-conditioning mode, the system kept the room temperature at 23°C while the relative humidity fluctuated between 60 and 70% RH. The RH fluctuation was the first sign that there was a problem with the serviced HVAC system, as the thermostat was set for summer conditions of 26°C and 55% RH. The cause was later determined to originate from a wall-mounted thermostat, which was not calibrated over five years, and had since degraded in performance. Further, it was determined that after the retrofit, the thermostat was not sending correct RH data to the HVAC handling unit, resulting in no RH control. Also, occupants complained of low temperature as they were dressed for light summer work. Air velocity did not exceed 0.3 m/s in any field study, and MRT had little to no offset from room temperature.

The results from FS2 showed that thermal comfort parameters could not be kept constant throughout the measurement period and that the HVAC system was unable to compensate for the heat gains through the glazing. From around 13:00 h the room conditions were influenced by direct sunlight, and the air temperature rose to 27.5°C while the MRT reached 29.5°C . Also, as air temperature rose, the drop in RH was observed from a setpoint of 55%. The draft was not present.

FS3 had the most stable thermal conditions. However, the room air temperature was high at around 28°C , which was 2°C higher than the setpoint. The MRT showed little variance from the dry bulb air temperature. The RH was stable at the setpoint of 50%.

As shown in [Fig. 5](#), continuous measurements showed excellent IAQ parameters in FS1. This is to be expected as the HVAC system has a volumetric airflow of fresh air that exceeds 30 l/s per occupant, previously designed to be a printing room. The CO_2 levels are nearly the same

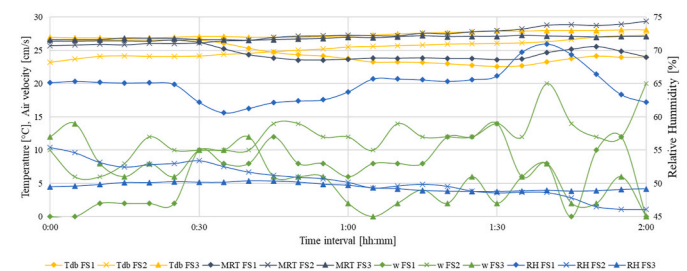


Fig. 4. 5 min averaged thermal comfort parameters for all field studies. The measurements took place on different dates and times, but the time intervals were matched for the purpose of result comparison.

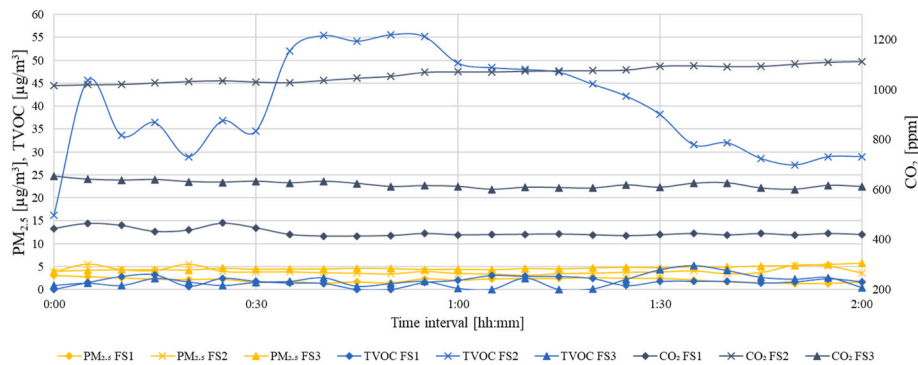


Fig. 5. 5 min averaged IAQ parameters for all field studies.

as outdoors. The same was true for PM_{2.5} and TVOC, as they had low concentrations throughout the measurements.

For FS2, it was expected to record higher TVOC levels as a result of a direct connection between the measured space and the printing room. The TVOC concentrations peaked at 57 μg/m³; however, the concentration was generally low and not significant enough to impact the IAQ component. The PM_{2.5} concentrations were also low, while the CO₂ concentrations continuously rose during the study, peaking at 1100 ppm. The results indicate that the sizing of the ventilation system was inadequate for maintaining the CO₂ levels below 1000 ppm as recommended by the standards. Excellent IAQ conditions can be observed in the FS3. The CO₂, PM_{2.5}, and TVOC concentrations were low and stable with full occupancy.

The leading cause for background noise in FS1 was attributed to the HVAC system, which, as mentioned before, was converted from industrial use. Before the retrofit, the background noise was 78 dB. After the retrofit of duct silencers, the noise was reduced to 53 dB, which was confirmed with the measurements (the unit turns on at the 15-min mark). The little spike in noise is attributed to an industrial refrigeration compressor. Because the occupants primarily did computer work, lighting was not given much attention. The building had sufficient artificial lighting, which was not used. The daylighting resulted in ~250 lux at table level, whereas with the artificial lighting turned on, the illumination at desk level reached ~500 lux, as can be seen for FS1 in Fig. 6.

The illuminance levels in FS2 were a strong function of daylighting. The drop in the lux level from ~500 lux to 270 lux was evident when the shading was used. The background noise is mainly contributed to the office work at around 50 dB, as the office is professionally sound insulated from the printing room.

The illumination was constant through the measurement period at ~345 lux. The room was equipped with noise-dampening wall panels to reduce the echo. As the occupancy reduced, so did the background noise from 45 to 40 dB. Such low background noise is to be expected in the educational facilities.

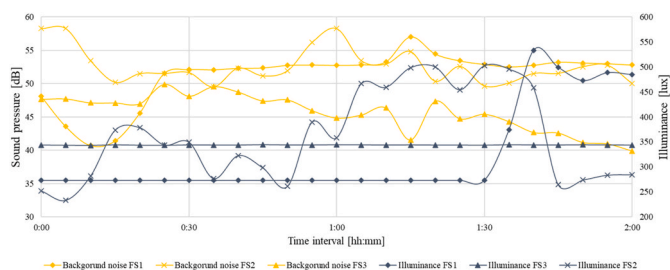


Fig. 6. 5 min averaged illumination and background noise level for all field studies.

4.2. Summary of IEQ indexes

Fig. 7 shows the results of the continuous IEQ index component calculation in the three field study buildings. Generally, the IAQ components of the IEQ index were ranked the highest in all buildings, achieving category *I* in the FS1 and FS3 while FS2 scored in the *II* category. Thermal comfort proved to be the most negatively affected component in FS2 and FS3. In FS3 it teetered between *II* and *III* categories, and in FS2 it degraded from *II* category to a practically intolerable thermal environment to occupants caused by direct solar radiation through the fenestration.

The visual comfort component was consistent in all three field studies and in *II* category (FS1 achieved *I* category after the artificial lights were turned on by researchers). The FS2 shows a good example of a high daylighting impact on the index, with the most varying value in time.

FS2 proved to have the best acoustics component, only departing from *I* category for a short period of time. The noise of the ventilation system kept the FS1 at constant *II* category, while in the FS2 it fluctuated depending on the office activities.

Fig. 8 presents the time evolution of the overall IEQ index. The results from FS1 show that the conditions were maintained at high quality and not falling from the *I* category mainly attributed to IAQ and thermal comfort. The results from the FS2 indicate that an open office continuously achieved *II* category with the IAQ, acoustic and visual component offsetting the thermal comfort negative impact. The mean absolute error between the average survey results and the overall IEQ index for the three field studies is <3%, suggesting that the platform can be used for generating large IEQ datasets for researchers and practitioners.

In FS3, thermal comfort was also the component that influenced the IEQ results the most. It is the main reason FS3 continuously fell into the *II* category. The thermal environment conditions were the most stable out of all of the experiments, even with the continuous change in occupancy. *I* category would be achieved if a simple step of turning down the thermostat by 2 °C were to be implemented.

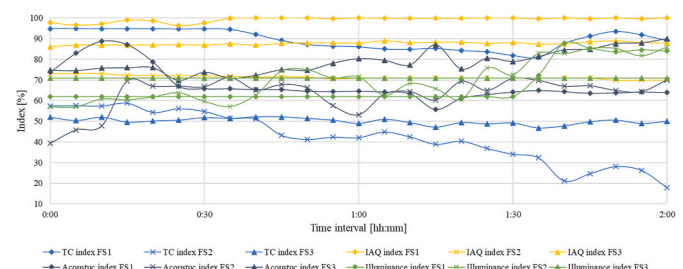


Fig. 7. 5 min averaged IEQ index components for all field studies.

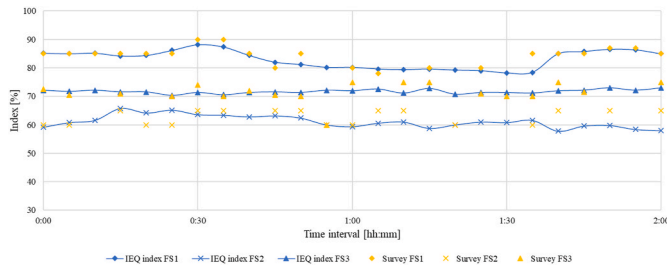


Fig. 8. 5 min averaged real-time IEQ index measurement with 5 min averaged survey results for all field studies.

4.3. Survey-based weighting coefficients

In addition to the overall IEQ index, Fig. 8 also presents the results of occupancy IEQ surveys. As seen, a close agreement can be observed between the I_{IEQ} and the surveyed IEQ. Firstly, similar to the previously mentioned studies, the results from subjective survey evaluation indicate that indoor air quality was subjectively the most critical IEQ component to occupants in both office buildings, with $W_{IAQ_{FS1}} = 0.33$ ($p = 0.001$), $W_{IAQ_{FS2}} = 0.37$ ($p = 0.002$), followed by thermal comfort $W_{TC_{FS1}} = 0.29$ ($p = 0.005$), $W_{TC_{FS2}} = 0.28$ ($p = 0.007$). The acoustic comfort $W_{AC_{FS1}} = 0.22$ ($p = 0.03$), $W_{AC_{FS2}} = 0.17$ ($p = 0.07$) and illumination $W_{IL_{FS1}} = 0.16$ ($p = 0.17$), $W_{IL_{FS2}} = 0.18$ ($p = 0.21$) were rated to have less contribution and were rated in third or fourth place. Further, weighting coefficients were averaged for FS1 and FS2 giving the I_{IEQ} equation:

$$I_{IEQ_{average}} = 0.285 \cdot I_{TC} + 0.35 \cdot I_{IAQ} + 0.195 \cdot I_{AC} + 0.17 \cdot I_{IL} \quad (12)$$

When the regression was applied to the educational building (FS3), the coefficients differed ($R^2 = 93.1\%$ and $F = 0.0002$). In that case, thermal comfort was the most dominant component, with $W_{TC} = 0.31$ ($p = 0.00006$) followed by indoor air quality and acoustics, both with weighting at 0.25, and lighting with the least importance $W_{IL} = 0.19$ ($p = 0.33$). Equation (1) has a shape:

$$I_{IEQ_{FS3}} = 0.31 \cdot I_{TC} + 0.25 \cdot I_{IAQ} + 0.25 \cdot I_{AC} + 0.19 \cdot I_{IL} \quad (14)$$

The Equation now falls better in line with the proposed “crude” scheme (Piasecki et al., 2017) in which all IEQ components matter equally. The differences in the results between office and educational buildings imply that occupants have different IEQ preferences depending on the indoor environment and their activities.

4.4. Research limitations

First, the authors acknowledge that the equations used to determine the IEQ components and the IEQ index from the measured parameters are not exclusive and that suitable alternatives can be found in the literature. Measurements were limited to several hours only, and the future investigations to cover longer time intervals. Also, all of the results were obtained during the cooling season and need to be verified in the heating HVAC conditions. The measurements in the three case study buildings were not conducted on the same day (but rather one month apart), which could introduce additional bias in the results. Further, the device was placed at a single location, in the center of the room to measure average parameters. However, conditions in large open offices cannot be always considered homogeneous. Several devices should be placed to cover specific floor areas, and the index should be averaged to get the most accuracy.

Currently, the platform adopts the assumption of the use of static clothing insulation (surface skin temperature and clothing surface temperature) and the static metabolic rate of occupants. Further development of the platform will explore the possibility of the integration of smart wearable sensors for tracking these parameters. Nizetić

et al. (2020) showed that these devices have the potential to accurately gather occupants’ metabolic rate and surface skin temperature. Their study also showed that, regardless of the season, activity levels dynamically varied between 1.0 and 2.0 Met, which is above the static values prescribed in ASHRAE and ISO standards.

Even though the models show good agreement with the past research and field data, more buildings (in number and in type) with larger survey respondents pool is needed for the results to be accepted on a general level. Also, current results are representative of temperate climates with warm summers and are specific to the office and educational space types. Despite the limitations, this study demonstrates a successful development and deployment of a new low-cost IEQ monitoring platform and offers unique insights that support further refinement of a unified IEQ index.

5. Conclusions

Currently, researchers, practitioners, and green building certification program developers are aspiring to develop and implement new metrics and tools for IEQ assessment to enhance human health, productivity, and well-being. The new tools must be continuous, affordable, accurate, reliable, intuitive, and non-intrusive to the occupants. Recent years have witnessed an influx of low-cost IEQ monitors on the market, which opened up an opportunity for building managers and occupants to receive ubiquitous information about their environment and act upon it. Yet, the majority of existing IEQ monitoring platforms are still unable to assess indoor environments holistically.

In this study, the research methodology for the development of a low-cost continuous IEQ sensing platform “ENVIRA” was presented, followed by its deployment in three case study buildings (two offices and one educational building). Through a combination of continuous measurements of individual IEQ parameters (thermal comfort, indoor air quality, lighting, and acoustics) and survey-based occupant IEQ perceptions data, this study provides new results of the importance of the individual parameters used to assess overall IEQ performance. Total of 125 occupants participated in the field studies – 69 in two open offices and 56 in the educational building. The results from occupant survey regression analysis show that occupants had similar preferences in office-type buildings. When the results are averaged for both buildings, indoor air quality is observed as the most critical environmental parameter with the weight of 0.35 followed by thermal comfort (0.285) acoustics (0.195), and illumination (0.17). The preference of occupants differed in the educational building where thermal comfort was seen as most important (0.31) followed by indoor air quality and acoustics equally (0.25), and lighting as the least important (0.19). Calculated overall IEQ index and occupant perception results showed good agreement with mean absolute error of <3%. At present, some green building certification programs assume equal importance among the four IEQ categories. Further research is needed to understand the weight of individual IEQ components, especially in relation to different building typologies.

The criteria used for indoor environments significantly influence the energy consumption of building design and operation stages. Occupants that are uncomfortable with their indoor environment are prone to take steps in achieving comfort which may have further energy implications. In addition to continuous assessment of IEQ parameters, the ENVIRA platform has demonstrated the potential to identify the HVAC system’s shortfalls, which could lead to improved system maintenance and reduction of the building’s energy consumption. Thereby, the proposed sensing platform shows potential to be a robust tool for energy-efficient management of HVAC systems in a data-driven manner. In the near future, the ENVIRA IEQ platform will be adapted for the use of the Zigbee mesh network, which is currently favored within the building management systems (BMS) industry. Thus, it will be capable of BMS integration. Further, the platform will be developed to support multiple devices with parallel data streams enabling complex spatiotemporal

analyses.

CRedit authorship contribution statement

Igor Mujan: Investigation, Methodology, Resources, Visualization, Writing – original draft, Writing – review & editing. **Dusan Licina:** Methodology, Writing – original draft, Supervision. **Miroslav Kljajić:** Investigation, Methodology, Resources. **Ana Čulić:** Investigation, Resources. **Aleksandar S. Anđelković:** Supervision, Methodology, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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